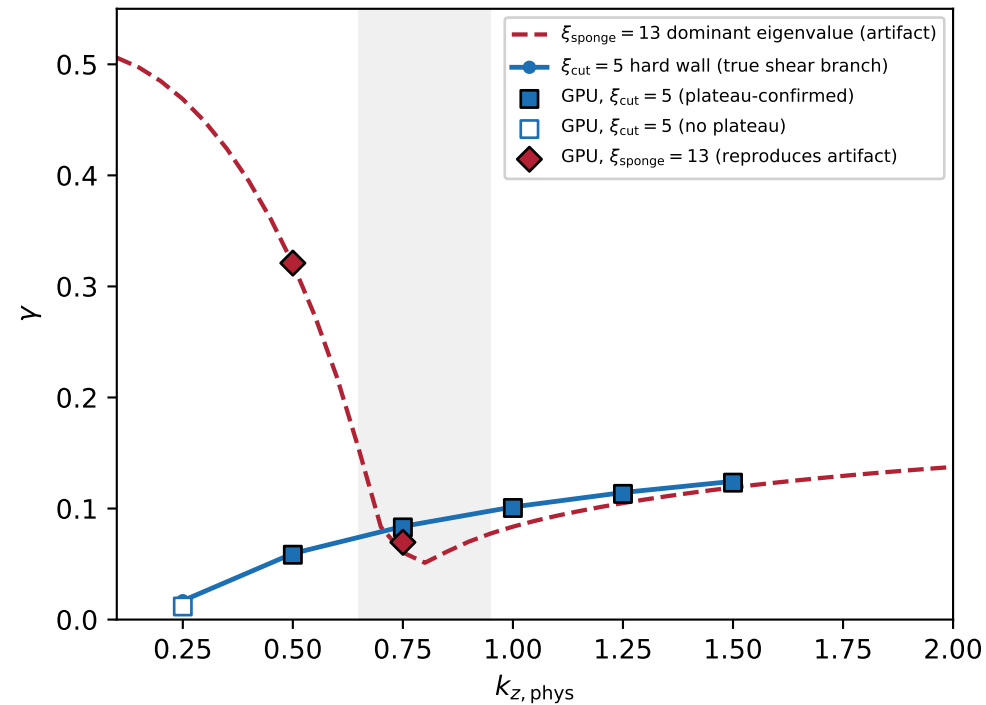
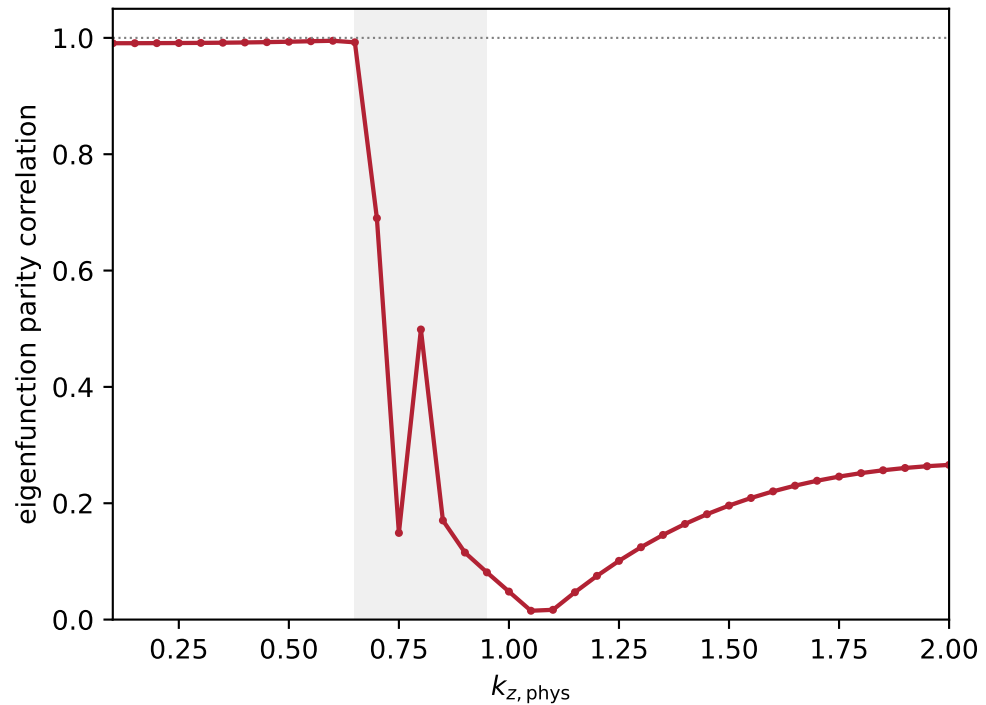


$\alpha = 2, V_0 = 0.05, \varepsilon_s = 0.15$



coherence collapses exactly at the ``dip''



Shaded band: where the $\xi_{\text{sponge}} = 13$ branch is pinned near the sponge wall ($\xi_{\text{peak}} \approx \xi_{\text{sponge}}$, A_z/B_y blows up 2.6→40--150).